ISEA INTERNSHIP

Raw Socket Packet Analysis and Protocol Investigation

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1. Objective

The objective of this assignment is to understand raw socket programming, IP packet structure, protocol headers, and packet analysis using Linux and Wireshark. Students will write a C program using raw sockets, generate

protocol-specific traffic, capture packets, compare results with Wireshark, and analyze packet header fields.

2.System Configuration

Operating System → **Ubuntu**

GCC Compiler →**gcc version 13.x**

Network Interface→l**o (Loopback Interface)**

3. Compile and Run Commands

Compile commands : *gcc raw_capture.c -o raw_capture*

Run commands :*sudo ./raw_capture*

4. Program Design

The raw socket program was implemented using:

socket(AF_INET, SOCK_RAW, IPPROTO_TCP)

The program captures TCP packets and extracts information from the IP and TCP headers.

IP Header Fields Captured:

- Source IP Address
- Destination IP Address
- Protocol Number
- TTL Value
- Packet Size

TCP Header Fields Captured:

- Source Port
- Destination Port
- TCP Flags

The program captures and displays 20 TCP packets.

5. Traffic Generation

Terminal 1 (Server): *nc -l 5000*

Terminal 2 (Client): *nc 127.0.0.1 5000*

Message sent: Hello Kaushal

Captured Interface: **lo (Loopback Interface)**

6. Experimental Results

Sample output from the raw socket program:

ROLL_NO=CSB24069

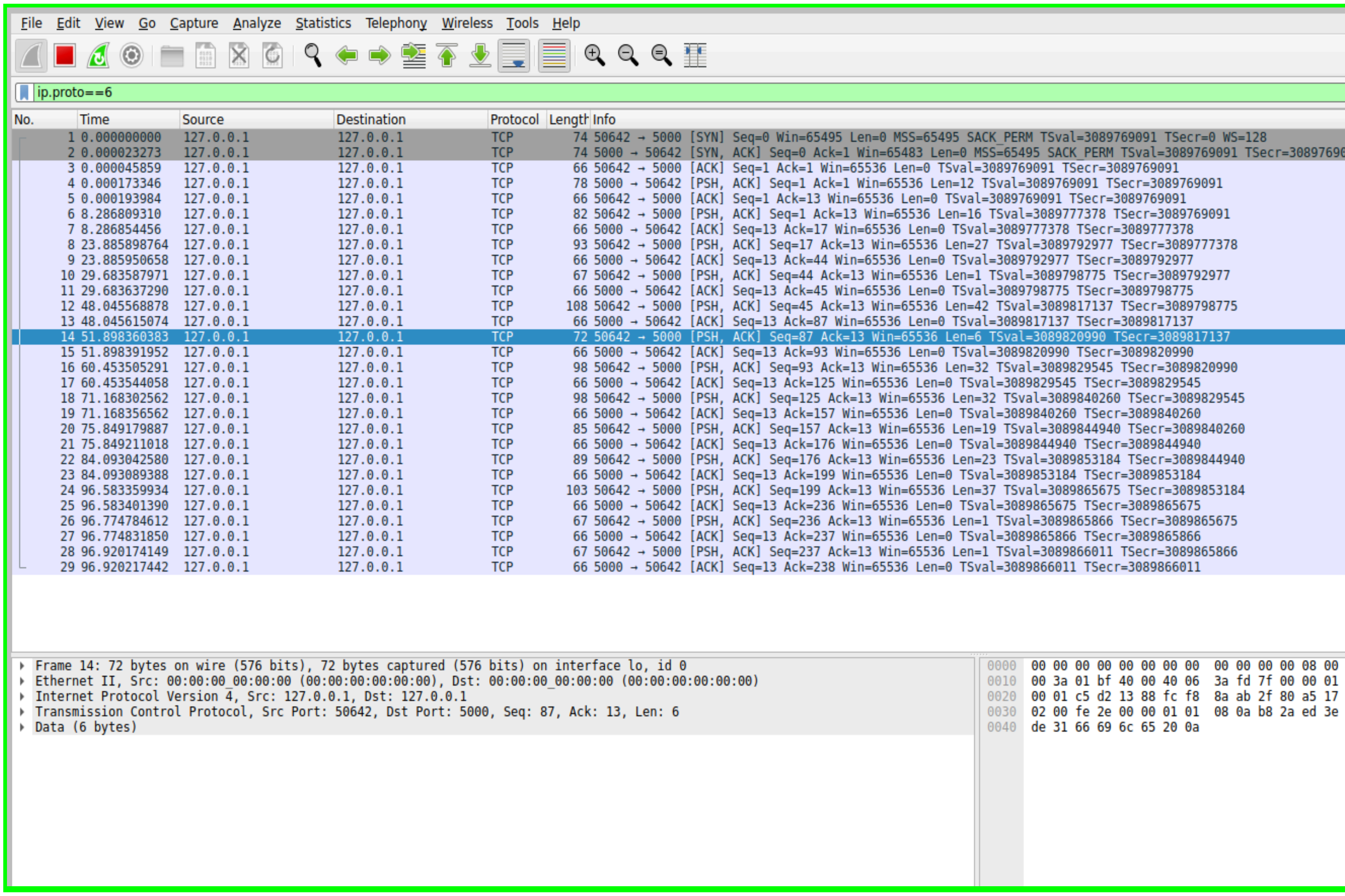
ASSIGNED_PROTOCOL=TCP

```
PACKET_NO=20
SRC_IP=142.251.155.119
DST_IP=172.29.115.55
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=117
PACKET_SIZE=1320
SRC_PORT=443
DST_PORT=39458
TCP_FLAGS=ACK PSH
```

Similarly, the program captured a total of 20 TCP packets.

7. Wireshark Verification

Display filter used: ip.proto == 6



The values observed in Wireshark matched the information displayed by the raw socket program.

8. Header Analysis

1. Why is the TTL value not zero?

TTL (Time To Live) represents the maximum number of hops a packet can travel through a network. Every router decreases the TTL value by one. The TTL is not zero because the packet reached the destination before exceeding its hop limit. If TTL becomes zero, the packet is discarded and an ICMP Time Exceeded message is generated.

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2. Why do packet sizes differ?

Packet sizes vary because different TCP packets contain different amounts of data. SYN and ACK packets usually contain only header information, while data packets contain additional application data, increasing the total packet size.

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3. What information does the protocol field provide?

The protocol field in the IP header identifies the transport layer protocol encapsulated inside the IP packet.

Examples:

- ICMP → Protocol Number 1
- TCP → Protocol Number 6
- UDP → Protocol Number 17

In this experiment, the protocol number was 6, indicating TCP packets.

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4. What could happen if the protocol field is modified?

If the protocol field is changed, the receiving device may process the packet using the wrong protocol handler or discard it completely. This can cause communication failures or incorrect interpretation of the packet.

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5. What do the TCP-specific fields reveal?

The TCP source and destination ports identify the applications involved in communication.

TCP flags indicate the current state of the connection:

- SYN: Starts a connection
- SYN-ACK: Accepts a connection request
- ACK: Acknowledges received data
- PSH: Transfers application data immediately
- FIN: Terminates a connection

9. Program Enhancement

Selected Field: IP Version4

Purpose

The IP version field specifies the version of the Internet Protocol used by the packet.

```
ROLL_NO=CSB24069
ASSIGNED_PROTOCOL=TCP

PACKET_NO=1
SRC_IP=127.0.0.1
DST_IP=127.0.0.1
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=64
PACKET_SIZE=60
IP_HEADER_LENGTH=20 bytes
SRC_PORT=57212
DST_PORT=5000
TCP_FLAGS=SYN
PAYLOAD_SIZE=0 bytes

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PACKET_NO=2
SRC_IP=127.0.0.1
DST_IP=127.0.0.1
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=64
PACKET_SIZE=60
IP_HEADER_LENGTH=20 bytes
SRC_PORT=5000
DST_PORT=57212
TCP_FLAGS=SYN ACK
PAYLOAD_SIZE=0 bytes

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PACKET_NO=3
SRC_IP=127.0.0.1
DST_IP=127.0.0.1
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=64
PACKET_SIZE=52
IP_HEADER_LENGTH=20 bytes
SRC_PORT=57212
DST_PORT=5000
TCP_FLAGS=ACK
PAYLOAD_SIZE=0 bytes

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PACKET_NO=4
SRC_IP=127.0.0.1
DST_IP=127.0.0.1
PROTOCOL=TCP
PROTOCOL_NO=6
TTL=64
PACKET_SIZE=66
IP_HEADER_LENGTH=20 bytes
SRC_PORT=57212
DST_PORT=5000
TCP_FLAGS=ACK PSH
PAYLOAD_SIZE=14 bytes
PAYLOAD_DATA=hello kaushal.

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```

Observation

The IP header length remained constant at 20 bytes, indicating a standard IPv4 header without optional fields.

10. Reflection Answers

1. Why are root privileges required for raw sockets?

Raw sockets provide direct access to network packets, including header information. Root privileges are required to prevent unauthorized users from monitoring or modifying network traffic.

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2. How are raw sockets different from TCP/UDP sockets?

Raw sockets provide direct access to complete packets and allow the user to examine IP and transport layer headers. TCP/UDP sockets provide only the application data, while the operating system handles the protocol details.

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3. One advantage and one limitation of raw sockets

Advantage:
Raw sockets allow detailed packet analysis and custom packet creation.

Limitation:
They require administrative privileges and are more complex to program.

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4. One networking or cybersecurity application of raw sockets

Raw sockets are used in packet sniffers, network analyzers, intrusion detection systems (IDS), and security monitoring tools.

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11. Conclusion

This experiment demonstrated the use of raw sockets for capturing TCP packets and analyzing IP and TCP header fields. The captured packet details were successfully verified using Wireshark, and the results from the raw socket program matched the information observed in the packet analyzer.
